

The Inflationary Footprint of Fiscal Policy: Evidence from Disaggregated U.S. Spending*

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Abstract

Theory shows that fiscal spending shocks generate wealth effects when they are perceived to lack credible backing. Fiscal backing is an intertemporal concept, as it depends on expectations about the availability of fiscal resources to cover debt over time, making it difficult to measure empirically. For defense spending, immediate fund availability serves as a useful proxy, whereas for mandatory programs it does not, and mandatory spending does not generate wealth effects except when it entails a persistent structural shift, as in the Social Security expansions of the 1960s. Unbacked wartime buildups are a significant driver of inflation and output.

Keywords: Fiscal backing, wealth effects, Ricardianity, Monetary and fiscal interactions, anticipated shocks, fiscal shocks.

JEL classification: E52, E62, E63.

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1 Introduction

Not all government spending affects the economy in the same way. While fiscal expansions are often studied in aggregate, they encompass a wide variety of programs, ranging from entitlements such as Medicare and Social Security to discretionary items such as defense. In this paper, we investigate whether the effects of government spending on output, inflation, and interest rates vary across spending categories, and whether these effects depend on the degree of fiscal backing; that is, the extent to which spending is credibly supported by future fiscal surpluses.

According to the fiscal theory of the price level, a lack of fiscal backing turns fiscal shocks into wealth effects that generate inflationary pressures, which the central bank accommodates rather than offsets. Measuring fiscal backing is difficult, however, because it is an intertemporal concept that depends on expectations about the availability of fiscal resources over time. To address this challenge, we construct indicators that capture slack or strain in the resources available to finance each type of spending. These indicators serve as proxies for fiscal backing and allow us to distinguish between programs announced during periods of fiscal slack and those announced under fiscal strain.

Our results show that immediate fund availability is a useful proxy for defense, but not for mandatory programs. Mandatory spending does not generate wealth effects, except when it involves a persistent structural shift, such as the Social Security expansions of the 1960s. In sum, we find evidence of inflationary pressures accommodated by monetary policy only in response to announcements of increases in national defense spending and to President Lyndon Johnson’s Great Society initiative, which permanently expanded Social Security spending. We examine the inflationary consequences of defense spending during major U.S. military conflicts. Large, debt-financed buildups during these episodes have historically contributed to inflation, as unbacked defense expansions triggered significant upward pressure on prices – a phenomenon we dub “warflation.”

We proceed by first focusing on anticipated shocks to Medicare and Medicaid programs and National Defense spending, which are among the largest components of mandatory and discretionary spending, respectively. We depart from the usual measures of government

shocks and derive them directly from the U.S. Federal Budget by exploiting the government’s expenditure forecasts for each spending category.¹ We follow the methodology of [Forni and Gambetti \(2016\)](#) and calculate distinct time series of shocks for our relevant spending categories.

We study the effects of these shocks using local projections. Crucially, given that the varying degree of fiscal strain may influence the strength of fiscal backing, we make our local projections state-dependent. We measure fiscal strain for Medicare and Medicaid spending by introducing a binary variable that indicates whether the combined funds – i.e., Hospital Insurance (HI) Trust Fund and Supplementary Medical Insurance (SMI) Trust Fund – are in deficit. These funds are intended to finance health care programs.² As a proxy for fiscal strain relative to National Defense spending, we use the Fiscal democracy Index. Developed by [Steuerle \(2010\)](#), this index measures the percentage of federal tax revenues that are not allocated to mandatory spending programs, indicating the portion of tax revenues available for discretionary spending over time. A lower value of this index suggests fewer resources available to the federal government to support increases in discretionary spending, which includes National Defense spending. The index dropped significantly from 65% in 1962 to well below zero in 2009, with two notable periods of decline: from 1969 to 1976 (though remaining positive) and from 2009 onward (becoming negative). We refer to these intervals of steep decline in the Fiscal democracy Index as periods of fiscal strain.

This analysis reveals that in periods of limited fiscal strain, both spending categories are expected to be supported by future fiscal adjustments, resulting in Ricardian effects. However, in periods of elevated fiscal strain for these two categories, only National Defense spending appears to trigger noticeable non-Ricardian effects. In contrast, shocks to Medicare and Medicaid spending generally exhibit Ricardian effects, regardless of the fiscal strain level, as indicated by the resources available in the HI and SMI funds.

We also look at the propagation of anticipated fiscal shocks to Social Security spending

¹Mandatory spending refers to expenditures required by law and generally not subject to annual review by Congress. Discretionary spending, by contrast, encompasses expenditures that Congress must review and approve annually through the appropriations process. These categories are discussed in more detail in [Section 2](#).

²Refer to [Section 3](#) for more information on the indices of fiscal restraint.

– another prominent category of mandatory spending in the U.S. Federal Budget. We measure fiscal strain for Social Security spending by introducing a binary variable that indicates whether both the Federal Old-Age and Survivors Insurance (OASI) and the Federal Disability Insurance (DI) programs are in deficit.

In line with health programs, the transmission of shocks to Social Security spending differs from that of National Defense, as its effects on economic activity are generally not statistically significant. However, when the funding allocated to Social Security is ample, anticipated increases in social spending tend to be more inflationary.

This finding can be interpreted as follows. Social security spending programs often create entitlements that are difficult to eliminate. These entitlements tend to be persistent, effectively permanent, or challenging to restructure – for example, pensions that individuals have incorporated into their financial planning cannot be easily phased out, even if funding becomes constrained over time. Indeed, the government’s capacity to financially support these entitlements is subject to change over time, influenced by demographics, economic growth, and other factors that are ex-ante unknown. As a result, assessing the fiscal backing of these long-term programs is inherently more complex, and they inevitably carry a significant risk of eventually lacking sufficient funding.

To support this interpretation, we present two key findings. First, the largest Social Security shocks in our sample occurred during the 1960s and the first half of the 1970s, a period when OASI and DI funds were abundant. These shocks capture President Lyndon Johnson’s Great Society initiatives – an ambitious plan to expand social spending and entitlements over many years. During this time, the U.S. economy was growing rapidly, and the demographic dividend was substantial, creating favorable funding conditions for the government to expand social spending. Indeed, OASI and DI funds – our proxies for financial strain in social spending – were largely in surplus in that period. Second, after excluding these shocks, the effects of Social Security shocks are Ricardian, regardless of the availability of OASI and DI funds.

This suggests that the observed non-Ricardian effects of Social Security shocks are primarily driven by the sizable shocks concentrated in that earlier phase. During that period,

even though the federal government was expanding entitlements and OASI and DI funds were plentiful, there may have been an expectation that financing such expensive and long-lasting social programs solely through fiscal instruments could prove politically unfeasible in the long run. This finding, which supports the view that the non-Ricardian effects of Social Security spending were concentrated primarily in the 1960s and early 1970s, aligns with the narrative presented in [Bianchi and Ilut \(2017\)](#), [Bianchi and Melosi \(2022\)](#) and [Cochrane \(2022\)](#).

While fiscal backing is a useful indicator of whether government spending has Ricardian effects, other institutional features also seem to play a role – for example, whether the spending creates entitlements. Overall, our results show that the inflationary effects of unbacked fiscal policies are contingent on both government spending categories and the timing of interventions. Importantly, one key factor affecting the ultimate economic outcome is the timing of the fiscal policy announcement. As [Ascari et al. \(2023\)](#) show, the typical lag between the announcement and subsequent implementation can yield opposite effects of anticipated government spending shocks on major economic variables, depending on whether the expenditures are expected to be backed by taxes or not. In light of this, we look at the effects of anticipated government spending shocks in our analysis.

Our approach features three main novelties. First, we look at different spending categories. Although the effects of government expenditures are usually studied using aggregate measures, such as government consumption of goods and services and investments, or government transfers, we disaggregate federal expenditures in the Federal Budget of the United States into their respective spending categories. We consider the distinction, explicitly present in the budget since the 1990s but also derivable in earlier years, between mandatory and discretionary spending, and we focus on their main subcategories. Second, we depart from the usual measures of anticipated government shocks proposed in the literature (see [Ramey and Shapiro, 1998](#); [Forni and Gambetti, 2016](#); [Fisher and Peters, 2010](#)) and derive these shocks directly from the U.S. Federal budget by exploiting the government’s expenditure forecasts for each spending category. As we will explain later, this approach allows us to obtain a measure of the government spending shock that is both *anticipated*

and *disaggregated*. Finally, we consider the financing capabilities of these different spending categories within the federal budget. The idea is to condition our analysis on a proxy of expected fiscal backing of the different spending categories.

Literature Review. This paper is related to four different streams of literature. The first encompasses the broad research on monetary-fiscal policy interactions, with a particular focus on the Fiscal Theory of the Price Level (FTPL). Foundational theoretical contributions include the seminal works by [Sargent and Wallace \(1981\)](#), [Leeper \(1991\)](#), [Sims \(1994\)](#) and [Woodford \(2001\)](#), which formalize the conditions under which fiscal policy can be a key driver of the price level. More recent studies embed FTPL mechanisms in structural models to explore their implications for macroeconomic dynamics and policy design (see, among others, [Cochrane \(2023\)](#) and [Leeper and Leith \(2016\)](#)). Building on this framework, a growing number of empirical studies investigate the presence of different monetary-fiscal policy regimes in U.S. data (see, among the others, [Bianchi, 2012](#); [Bianchi and Ilut, 2017](#); [Ascari et al., 2020](#); [Bianchi and Melosi, 2014](#); [Jacobson et al., 2024](#)).

The second stream of literature deals with fiscal foresight and the identification of fiscal policy shocks. This literature examines the impact on the economy of fiscal policy in VAR models (e.g., see [Blanchard and Perotti, 2002](#); [Ramey, 2011](#); [Auerbach and Gorodnichenko, 2012](#); [Forni and Gambetti, 2016](#); [Ramey and Shapiro, 1998](#); [Mertens and Ravn, 2011, 2012](#); [Ben Zeev and Pappa, 2017](#); [Fisher and Peters, 2010](#)) and stresses the importance of accurately distinguishing between anticipated and unanticipated government shocks. Legislative and implementation lags imply that private agents can adjust their behavior during the time between the announcement of a fiscal measure and its actuation. In fact, those who analyze the effects of an anticipated increase in taxes (see [Mertens and Ravn, 2012, 2011](#)) find that the production increases during the anticipation period and, as happens for unanticipated shocks, decreases once the tax increase is implemented. [Ramey \(2011\)](#), who analyses the effects of an anticipated government spending shock, finds that such a shock increases output but decreases consumption.

A relevant paper for our analysis is [Ascari et al. \(2023\)](#). They extend the results by [Beck-Friis and Willems \(2017\)](#), who study unanticipated changes in government expendi-

tures under a backed monetary regime and an unbacked fiscal regime, to consider anticipated public spending shocks. Theoretically, they show that fiscal foresight leads anticipated government spending shocks to have very different effects if one considers a monetary rather than a fiscal regime. In the monetary regime, increases in government expenditure are assumed to be fiscally backed. Following an anticipated spending shock, output, consumption, investment, and inflation decrease while real interest rates and debt increase. These variables show an opposite reaction in the fiscal regime, under unbacked fiscal expansions, as consumption increases through the expectation of positive wealth effects. Anticipated shocks can thus provide a potential identification mechanism in the data to distinguish these two regimes. They thus look at two sample periods that have been identified in the literature as a monetary and a fiscal regime and find that anticipated government spending shocks have very different effects during the anticipation period, as theory would predict: in the monetary regime they are contractionary; in the fiscal regime they are expansionary.

The third stream of literature is the one on the state dependence of spending multipliers (see [Auerbach and Gorodnichenko, 2012](#); [Ramey and Zubairy, 2018](#); [Caggiano et al., 2015](#)). Unlike these studies, however, our focus is not on fiscal multipliers per se. Rather, we are interested in the broader macroeconomic consequences of spending shocks, following the tradition of research inspired by the FTPL, such as [Leeper et al. \(2017\)](#), who emphasizes the central role of the policy regime in determining the size and sign of fiscal multipliers. In particular, our analysis, which examines anticipated federal spending shocks while conditioning on a proxy for the expected fiscal backing of different spending categories, is closely related to [Ascari et al. \(2024\)](#). They show that the effects of anticipated government spending shocks critically hinge on the state of tax expectations – that is, whether agents expect taxes to increase or decrease.

A fourth and relatively recent strand of the literature emphasizes the heterogeneity of public expenditure. [Cox et al. \(2024\)](#), for example, look at government spending not as a single large transaction, “one big G”, but as the aggregation of numerous smaller transactions, examining federal government procurement contracts through this lens. While we draw inspiration from their granular perspective, our approach is broader in scope. Rather than

focusing on a specific subset of spending, we consider a broader set of government expenditures, disaggregated in a manner consistent with how U.S. lawmakers structure the federal budget. This approach allows us to capture different dimensions of fiscal heterogeneity and offers a complementary angle to studies focused solely on procurement spending.

The remainder of the paper is structured as follows. Section 2 describes the data and the methodology used to construct our new measure of anticipated and disaggregated spending shocks. Section 3 presents the results. Section 4 concludes.

2 Data and Methodology

Rather than considering aggregate government expenditures, we disaggregate them according to the predominant type of spending: mandatory or discretionary. Mandatory spending refers to expenditures that are required by law and are generally not subject to annual review by Congress. It comprises entitlement programs that are difficult to modify or reduce, as their funding is set by eligibility criteria rather than through the annual appropriations process. Examples include Social Security, Medicare, and Medicaid, which offer guaranteed benefits to eligible individuals and are usually financed by payroll taxes.³ Discretionary spending, on the other hand, encompasses expenditures that Congress must review and approve annually through the appropriations process. This category covers a wide range of government programs, including National Defense (which constitutes about half of the total) and non-defense spending such as education, infrastructure, and many federal agency budgets. Congress decides the specific funding levels for each program within this category every fiscal year. It can be adjusted or reduced according to the policy priorities and available funds. In 2023, health care programs, primarily Medicare and Medicaid, constituted around 28% of the total budget. Defense spending, part of discretionary spending, received about 13-14% of the budget. The other large category of mandatory spending is Social Security, which received about 23% of the federal budget.

³A few mandatory programs, for example, veterans' disability compensation and pensions and Medicaid, are also funded through appropriation acts. While pure mandatory programs receive automatic funding, these *appropriated mandatory programs* require funds to be approved through an annual budget process, even though the provision of benefits remains mandatory.

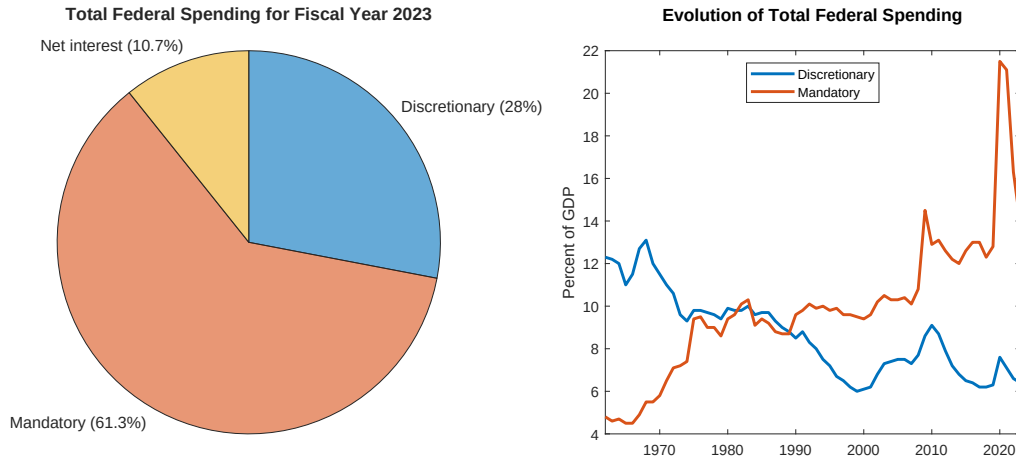


Figure 1: **Mandatory and discretionary spending.**

Left panel: composition of total federal spending in Fiscal Year 2023, highlighting the shares of mandatory, discretionary, and net interest spending. Right panel: evolution of yearly mandatory and discretionary spending as a share of GDP from 1962 to 2023.

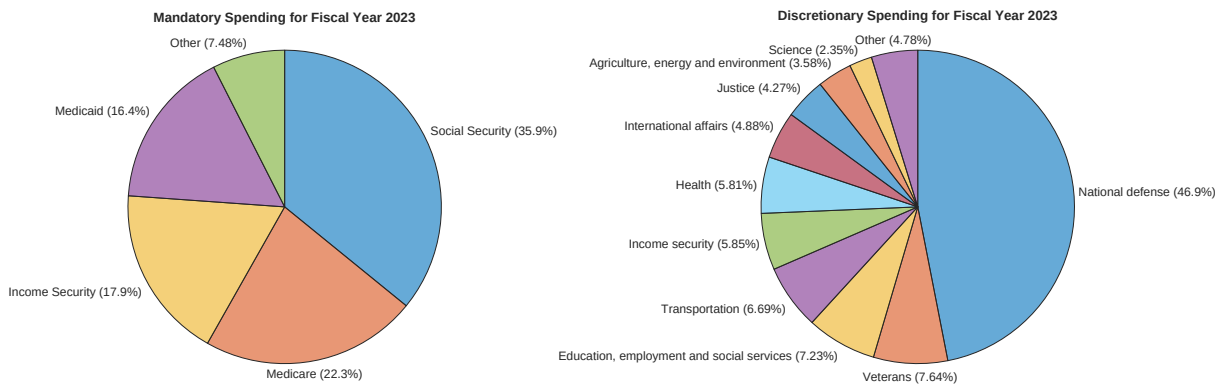


Figure 2: **Composition of mandatory and discretionary spending in Fiscal Year 2023.**

Left panel: breakdown of mandatory spending. Right panel: breakdown of discretionary spending. Mandatory spending is largely driven by Social Security, Medicare, and income security programs, while discretionary spending is dominated by national defense.

The left panel of Figure 1 shows the total federal spending data for Fiscal Year 2023. Mandatory spending is more than double the discretionary spending. This has not always been the case: the right panel shows how mandatory and discretionary spending have evolved since the 1960s. Note that, before the 1990s, mandatory spending was lower than discretionary spending.

Figure 2 shows the composition of mandatory and discretionary spending in 2023, respectively. Among the mandatory spending categories, Social Security (SS) has the largest

share at 36%, followed by Medicare and Medicaid (MM) at 22.3% and 16.4%, respectively. In contrast, slightly half of discretionary spending is allocated to National Defense (ND) expenditures, with the remainder going towards non-defense spending (education, transportation, environmental programs, and R&D).⁴

We use data from the U.S. federal budget to construct a measure of government spending shocks. The fiscal year runs from October 1 to September 30 of the following year and it is named after the calendar year in which it ends.⁵ The U.S. Federal Budget for a given fiscal year is typically released in March/April of the previous year. Because the budget is released before the start of the fiscal year, many of the data points and financial projections it contains are estimates. We exploit these estimates to construct our anticipated measure of government spending shock.

Following [Forni and Gambetti \(2016\)](#), we construct a quarterly measure of the one-year-ahead expected growth rate in real spending by subcategory. For instance, for Defense spending in 2022Q1, the measure is defined as:

$$f_{2022q1}^{\text{ND}} = \log \left\{ \frac{\mathbb{E}_{2022q1}(\text{real defense spending in 2023})}{\mathbb{E}_{2022q1}(\text{real defense spending in 2022})} \right\}$$

This measure is non-zero only in the quarter in which the budget is released; it is set to zero in all other quarters. We construct such a measure for the two broad spending categories (mandatory and discretionary), as well as for key subcategories such as Medicare and Medicaid combined, National Defense, and Social Security.

An important caveat must be addressed here: projected government spending may adjust to expected macroeconomic conditions, which would render our measure of fiscal shocks endogenous. To mitigate this concern, we adopt two alternative strategies. First, still taking National Defense as an example, we regress f_t^{ND} on the projected growth rate of real GDP reported in the budgets, and denote the residuals of this regression as $f_t^{\text{ND, purged}}$. Second, we exploit information from the Survey of Professional Forecasters (SPF), regressing $f_t^{\text{ND, purged}}$ on the expected growth rate of GDP and the expected change in unemployment,

⁴See [Appendix E](#) for figures showing the components of mandatory and discretionary spending, both in levels and growth rates, since the 1970s.

⁵For example, fiscal year 2025 begins on October 1, 2024, and ends on September 30, 2025.

and denote the corresponding residuals as $f_t^{\text{ND, SPF-purged}}$. The same procedure is applied to the fiscal shocks for each subcategory of spending. The time series of both the original and the purged fiscal shocks for all spending subcategories are reported in Appendix D. The series are highly similar, and the correlation between the original and purged shocks is consistently very high (always above 0.95), supporting the view that unpurged shocks still capture the exogenous component of government spending and suggesting that the potential endogeneity problem is not particularly severe. In the main text, we therefore report only the impulse responses to the baseline unpurged shocks, while leaving those obtained from the purged shocks to Appendix D.⁶

We employ the measures of fiscal shocks within Jorda’s (2005) local projection framework to evaluate the macroeconomic effects of anticipated government spending, disaggregated by subcategory. More in detail, each subcategory-specific measure is used as the shock variable in a series of local projections regressions, allowing us to estimate impulse response functions for a range of macroeconomic outcomes.

We consider both linear and state-dependent local projections. In the linear case, the local projection for horizon h is specified as

$$y_{t+h} = \alpha_h + \beta_h x_t + \sum_{i=1}^p \gamma_i' \mathbf{w}_{t-i} + v_{t+h}, \quad (1)$$

where y_t is the outcome variable, x_t is the measure of disaggregated government spending shocks, and $\mathbf{w}_t$ is the set of control variables. The outcomes include quarterly NIPA variables such as aggregate federal government purchases, aggregate federal transfers, GDP, private consumption and private investment. Following Ramey (2016), we adopt the Gordon-Krenn (2010) transformation: all NIPA variables are first deflated by the GDP deflator and then divided by potential GDP, estimated as a quadratic trend fitted on real GDP. We also

⁶A second issue arises from the fact that budgets report spending projections only in nominal terms, requiring a deflator to convert them into real quantities. However, projections of the GDP deflator are available only from 1976 onwards. Consequently, we construct our baseline measure of fiscal shocks using the actual GDP deflator series. As a robustness check, we also generate an alternative fiscal shock series using an expected GDP deflator series that we build by combining soft information from the budgets (1963–1968), SPF forecasts (1969–1975), and the projected deflator reported in the budgets (1976 onwards). Results are highly consistent across specifications. Therefore, the main text only reports the impulse responses to our baseline measures of fiscal shocks, which rely on actual inflation. Results based on projected inflation are available from the authors upon request.

consider other variables such as inflation, the differential between the interest rate and output growth ($r - g$), and debt over GDP. The combined reactions of the interest-rate–growth differential ($r - g$), inflation, output, and the debt-to-GDP ratio to anticipated fiscal shocks reveal the type of monetary–fiscal coordination – indicating whether the regime is one of monetary or fiscal dominance.⁷ The set of controls always includes four lags of output, of the outcome variable, and of the shock, to account for the presence of autocorrelation on our measures of anticipated government spending. We also include lagged transfers when computing impulse responses to MM and SS shock, as these subcategories are classified in NIPA tables as government social benefits to persons. We instead include lagged government purchases when computing impulse responses to ND and discretionary spending shocks. Finally, we include both transfers and purchases in the case of mandatory spending shocks.⁸ Results are robust to including taxes as controls.

Following [Ramey and Zubairy \(2018\)](#), we also account for state dependence by incorporating several state variables in our local projection framework. These variables allow us to assess whether the impulse responses to an anticipated shock in a specific spending category vary depending on whether the fiscal capacity in that category is high or low. For the model with state dependence, we estimate a set of regressions such as

$$y_{t+h} = \mathcal{I}_{\{z_{t-1}=1\}} \left(\alpha_h^1 + \beta_h^1 x_t + \sum_{i=1}^p (\gamma_{h,i}^1)' \mathbf{w}_{t-i} \right) + \mathcal{I}_{\{z_{t-1}=0\}} \left(\alpha_h^0 + \beta_h^0 x_t + \sum_{i=1}^p (\gamma_{h,i}^0)' \mathbf{w}_{t-i} \right) + v_{t+h}, \quad (2)$$

where $\mathcal{I}$ is a dummy variable that indicates the state of the economy when the shock hits. In particular, coefficient β_h^1 is the impulse response at horizon h conditional on the subcategory-specific fiscal capacity being low ($z_t = 1$), signaling fiscal strain; conversely, coefficient β_h^0 is

⁷The ‘growth-augmented Taylor principle’ (see [Elfsbacka Schmöller and McClung, 2024](#)) claims that, to maintain economic stability under a debt-stabilizing fiscal policy, central banks must raise ($r - g$). This helps control inflation, as higher growth can contribute to inflationary pressures. In contrast, under a fiscal authority which does not stabilize debt, central banks can stabilize the public debt by letting $r - g$ fall with inflation.

⁸While most mandatory spending consists of transfers to individuals, a small share includes government purchases of services, such as administrative costs for programs like Medicare, which are not classified as transfers. Results are barely unchanged if we use just transfers.

the impulse response in the presence of high fiscal capacity ($z_t = 0$), that is, absent fiscal strain.

We estimate local projection coefficients using the smooth local projection methodology of [Barnichon and Brownlees \(2019\)](#), applied both in the linear and in the state-dependent case. In this approach, impulse response coefficients at different horizons are expressed as linear combinations of B-spline basis functions, and the corresponding coefficients are estimated through generalized ridge regression, with a penalty parameter governing the degree of shrinkage.⁹ As in [Barnichon and Brownlees \(2019\)](#), we construct confidence intervals for the impulse response coefficients, employing a modified [Newey and West \(1987\)](#) heteroskedasticity- and autocorrelation-consistent estimator, adjusted to account for the penalty parameter. As shown by [Li et al. \(2024\)](#), smooth local projections typically outperform least-squares local projections under mean squared error loss, owing to their substantially lower variance at the cost of a modest increase in bias.

Unless otherwise stated, impulse responses are computed on the sample 1968q1-2019q4.

3 Empirical Findings

Using linear local projections [\(1\)](#), we analyze the impulse responses over the full sample period, from 1968q1 to 2019q4, to an anticipated shock to Medicare & Medicaid and to Social Security – which together account for the largest component of mandatory spending – but even to National Defense, the largest component of discretionary spending. [Figure 3](#) presents the impulse responses to a shock to each of the three spending components. Estimates are obtained using linear local projections, and are normalized such that the peak response in government purchases plus transfers equals one. Following the approach in [Ramey \(2016\)](#), the graphs for NIPA variables should be interpreted as variations in dollars; for example, a one-dollar peak increase in government purchases plus transfers corresponds to a peak decrease in GDP of 1.15 dollars. The scales of the other graphs are in percentage points. We report 68% and 90% confidence bands based on Newey-West corrections of standard

⁹The shrinkage parameter is selected by k -fold cross-validation, following [Barnichon and Brownlees \(2019\)](#).

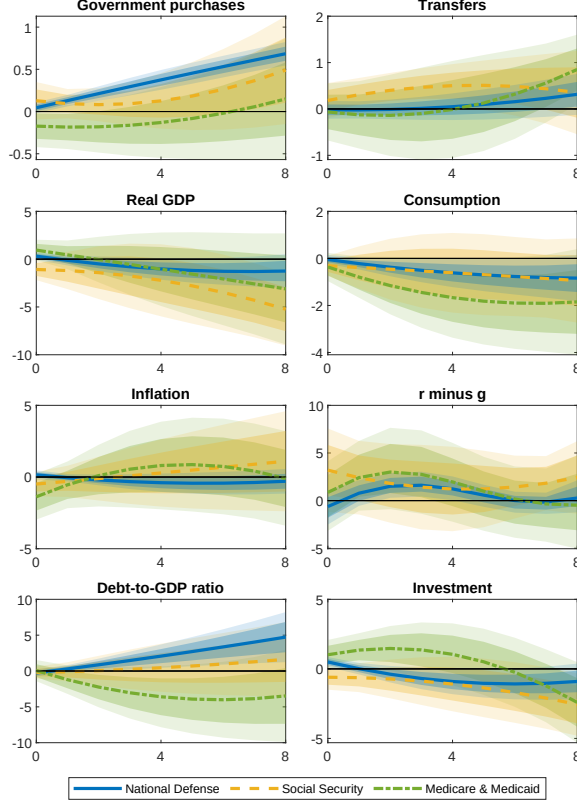


Figure 3: **Effects of anticipated spending shocks by category: Medicare & Medicaid, Social Security, and National Defense.**

Blue lines: responses to an anticipated shock to National Defense. Yellow lines: responses to an anticipated shock to Social Security. Green lines: responses to an anticipated shock to Medicare & Medicaid. The set of controls always includes four lags of the shock, output, and the outcome variable, plus four lags of transfers for panel Medicare & Medicaid and Social Security, or four lags of government purchases for National Defense. Each panel reports impulse responses from linear local projections estimated over the sample 1968q1-2019q4. Shaded areas: 68% (dark) and 90% (light) confidence bands.

errors. As the figure shows, one can not detect any statistically significant difference in the response of these components. Output, consumption and investment decrease following an anticipated shock while $(r - g)$ increases. A slightly different behavior can be seen in inflation, but again none of the impulse responses is statistically significant.

Fiscal backing. The responses to the three spending categories shown in Figure 3 are not conditioned on the availability of resources specific to each category. A key feature of our approach is the focus on anticipated shocks across spending categories, conditional on the availability of resources at the time of the announcement. This conditioning serves as a rough proxy for fiscal backing, in the sense of Bianchi et al. (2023). Launching a new

government program during a period of acute fiscal strain is more likely to be perceived as unbacked spending – an interpretation that, according to theory, critically shapes the program’s macroeconomic effects.

In what follows, we make our local projections dependent on the fiscal strain in the corresponding budget categories. Section 3.1 examines the effects of an anticipated shock to Medicare & Medicaid and National Defense spending, two sub-components of mandatory and discretionary spending, respectively. Section 3.2 analyses Social Security, the largest component of mandatory spending. While the main text concentrates on these key subcomponents, Appendix B extends the analysis to assess whether the results for these sub-components can be generalized to their aggregate categories of mandatory and discretionary spending. Finally, Section 3.3 takes stock of the results and provides a comparative overview, while also examining a special case within defense spending: the phenomenon of “warflation”, that is, inflation during major war events.

3.1 Medicare & Medicaid and National Defense

We begin by presenting the impulse response functions to an anticipated shock to Medicare & Medicaid and National Defense. We want to condition the analysis of the effects of these spending categories on the respective “fiscal capacity” of each category.

Notably, a deficit in the overall U.S. federal budget does not imply a deficit within each individual spending category. For instance, Medicare, funded through dedicated taxes channeled into trust funds, is considered to be in deficit when these trust funds face a shortfall. Conversely, National Defense, which falls under discretionary spending, is not tied to specific funding streams like those supporting mandatory programs. Its financing, therefore, depends on how Congress allocates the remaining tax revenues after mandatory expenditures have been covered, along with debt issuance. Therefore, to assess when the fiscal capacity of these components is under strain, we will employ two different measures, which we explain in turn below.

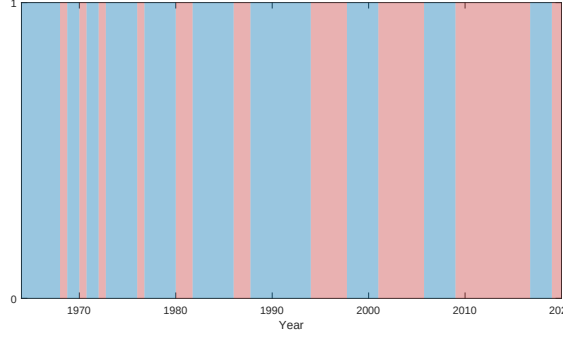


Figure 4: **Fiscal capacity for Medicare & Medicaid.**

Periods of constrained fiscal capacity to finance MM spending, based on the status of the Hospital Insurance trust fund and the Supplementary Medical Insurance trust fund. Shaded areas reflect the joint status of the trust funds: periods of fiscal strain when both funds are in deficit (red areas), versus periods without fiscal strain when at least one fund is in surplus (blue areas).

Identifying fiscal capacity for each spending category. We look at two large mandatory programs, Medicare and Medicaid, considering them together.¹⁰ Medicare has two trust funds: the Hospital Insurance (HI) and the Supplementary Medical Insurance (SMI). We focus on the combined programs (HISMI) and look at the deficit/surplus in HISMI. We assign a value of one if both are in deficit, otherwise, we assign a value of zero. Although Medicaid does not operate through dedicated trust funds, we use the financial position of Medicare’s trust funds as a proxy for overall fiscal strain, given that both programs are major federal health expenditures and draw on overlapping sources of general revenue. The red areas in Figure 4 indicate periods in which both funds are simultaneously in deficit, reflecting strained fiscal capacity to finance MM spending. In contrast, blue areas correspond to periods without fiscal strain, meaning that at least one of the two funds is not in deficit.¹¹

In the case of National Defense, we identify periods of scarce fiscal resources by looking at the so called fiscal democracy index (henceforth, Fiscal index or FI), introduced by [Steuerle \(2010\)](#). This index measures the share of federal tax revenues not allocated to mandatory spending programs, indicating the percentage of taxes available for discretionary spending over time.

$$\text{Fiscal index} \equiv 1 - \frac{\text{Mandatory spending} + \text{Interest}}{\text{Total revenues}}$$

¹⁰Results are barely unchanged if we consider them separately.

¹¹The index takes a value of one to identify the following periods as times of strain for MM spending: 1968, 1970, 1972, 1976, 1980-81, 1986-87, 1994-97, 2001-05, 2009-16, 2018-19. For more details on the MM (and SS) trust funds, including data sources, see Appendix A.

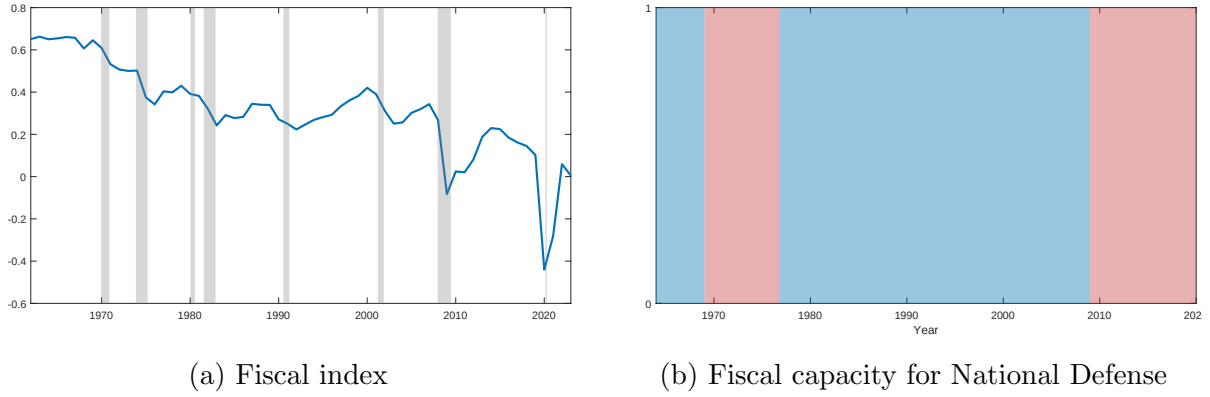


Figure 5: Fiscal index and fiscal capacity for National Defense spending.

Panel (a): evolution of Fiscal index. Note that the higher the index, the larger are funds available for discretionary spending and hence for ND spending. Grey bands indicate recessions. Panel (b): fiscal capacity for National Defense implied by the Fiscal index. Periods of fiscal strain are characterized by decreasing Fiscal index (in red), while periods of no fiscal strain are characterized by non-decreasing Fiscal index (in blue).

The higher the index, the stronger the fiscal democracy, as more funds are available for non-predetermined spending. As Figure 5a shows, over the past fifty years this index has dropped significantly, from 65% in 1962 to below zero in 2009. There were two notable declines: the first occurred between 1969 and 1976 (although the index remained positive), and the second began in 2009, when it turned negative.¹² We define periods of scarce fiscal resources for discretionary spending as those during which the Fiscal democracy Index experiences a marked decline, such as from 1969Q1 to 1976Q4 and from 2009Q1 onward. Red (blue) areas in Figure 5b identify periods with decreasing (non-decreasing) FI.

The results are robust even if one assigns a value of one only to the post-2009 period. This period, as Table 1 in Appendix D shows, is characterized by more frequent changes in the debt ceiling and numerous suspensions and, according to Bianchi et al. (2023), a resurgence of unfunded transfers spending.¹³

¹²“(...) in the year after the beginning of the current “financial crisis,” mandatory expenses including interest payments exceeded total government income. This was caused by a simultaneous decline in revenue and a rapid increase in mandatory spending. As a result, all discretionary spending, including military expenditure, had to be debt-financed. (...) The global financial crisis that began in 2008 has undone any gains in fiscal democracy made in the late 1990s and the mid-2000s. In effect, the crisis seems about to eliminate almost completely any space for fiscal discretion in the foreseeable future unless governments are willing, and able economically and legally, to take up new, additional debt.” (Streeck and Mertens, 2010)

¹³The debt ceiling limits total federal borrowing. Once reached, Congress must raise or suspend it to allow the Treasury to meet existing legal obligations.

Results. Figure 6 shows the results for state-dependent local projections, described by equation (2), for Medicare & Medicaid and National Defense in the period 1968q1-2019q4. Panel (a) presents impulse responses to an anticipated shock to Medicare & Medicaid. Given that this spending category largely consists of transfers to individuals, each local projection includes four lags of transfers, as well as four lags each of the shock, output, and the outcome variable. The red (blue) lines represent periods in which the HISMI trust funds are in deficit (surplus). Under both states, the initial response of transfers is relatively muted but subsequently increases, becoming statistically significant after two years. This pattern aligns with the interpretation that the shock signals news about a future rise in transfer spending. The impulse responses of macro variables are broadly similar across states. Hence, the lack of dedicated funding does not alter the propagation mechanism of anticipated shocks to MM spending, which remains consistent with Ricardian dynamics.

Panel (b) presents impulse responses to an anticipated shock to National Defense spending. In this scenario, we include four lags of government purchases instead of transfers, and condition on the presence of slack (red lines) or tightness (blue lines) in the Fiscal index. Our findings indicate a significant increase in government purchases under both fiscal states. On the other hand, the availability of resources to fund discretionary spending seems to play a crucial role in shaping the responses of real activity, inflation, and the debt-to-GDP ratio. Specifically, the responses of these variables are flipped in the two cases: when fiscal resources are abundant, the effects are muted and broadly consistent with Ricardian behavior; when resources are scarce, the shocks generate pronounced non-Ricardian effects. In the latter case, the anticipated increase in ND spending is more likely to be perceived as unbacked by future fiscal adjustment. This result aligns closely with theoretical predictions: Unbacked fiscal expansions are inflationary, stimulate economic activity and decrease the growth-interest rate differential, thus reducing the debt-to-GDP ratio.

In Appendix C, we assess the robustness of these findings across two distinct macroeconomic regimes. We split the sample in the Great Inflation and the ZLB period (1964q1-1979q2 and 2009q1-2015q4, respectively), and the Great Moderation years (1984q1-2008q4). The literature typically identifies the former as unbacked *fiscal regimes* and the latter as a

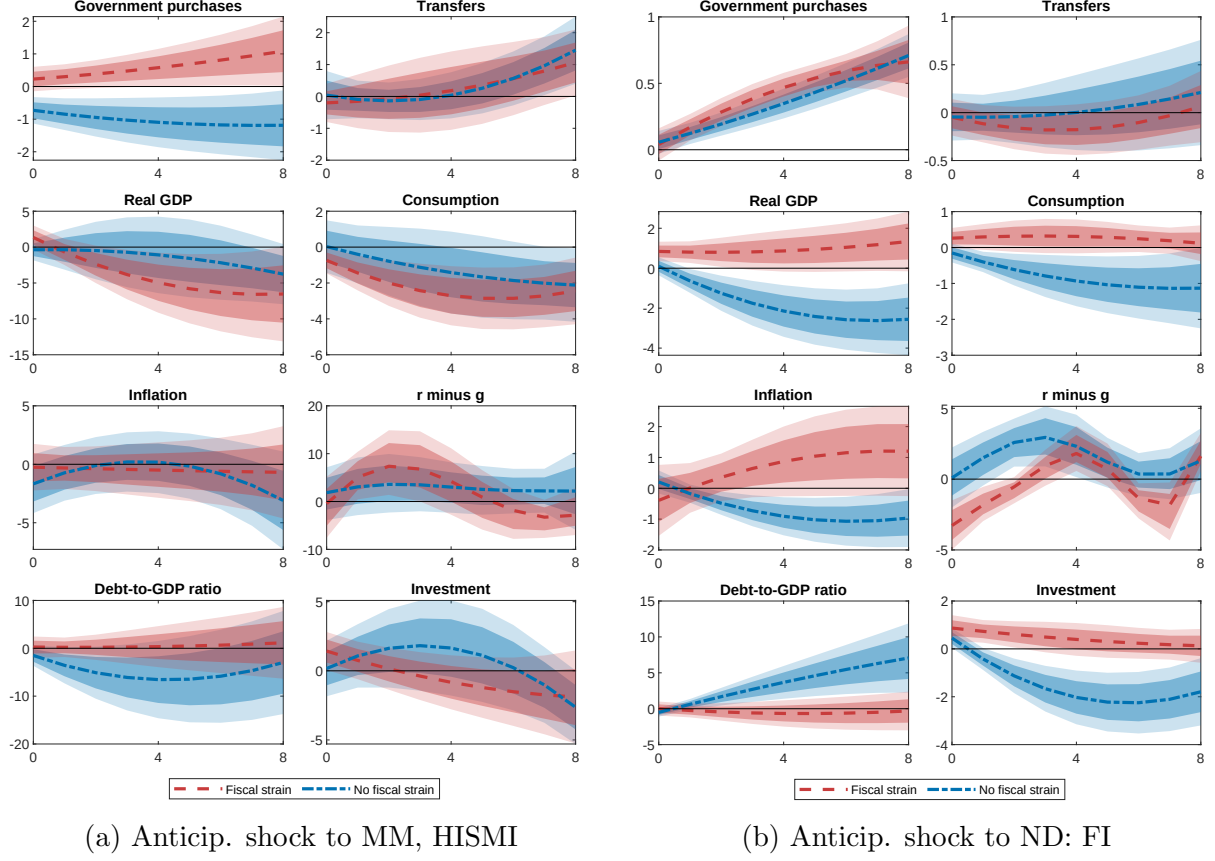


Figure 6: **Impulse responses to anticipated shocks to Medicare and Medicaid (MM) and National Defense (ND), conditional on the respective fiscal capacity.**

Panel (a): responses to an anticipated shock to Medicare & Medicaid, conditioning on the periods of fiscal strain (red lines) or absence of fiscal strain (blue lines) in the HISMI funds, as described in Figure 4. Panel (b): responses to an anticipated shock to National Defense, conditioning on the periods of fiscal strain (red lines) or absence of fiscal strain (blue lines) in the Fiscal index, as described in Figure 5b. Each panel reports impulse responses from state-dependent local projections estimated over the sample 1968q1-2019q4. The set of controls always includes four lags of the shock, output, and the outcome variable, plus four lags of transfers for panel (a) or government purchases for panel (b). Shaded areas: 68% (dark) and 90% (light) confidence bands.

backed *monetary regime*. We find that the effects of ND spending are Ricardian under the monetary regime and non-Ricardian under the fiscal regime. These findings are consistent with those in [Ascari et al. \(2023\)](#).¹⁴

3.2 Social Security

We now turn to the effects of anticipated changes in projected SS spending. As before, we begin by identifying indicators of slack or strain in the fiscal resources available to finance

¹⁴This is not surprising given that they primarily focus on defense spending shocks.

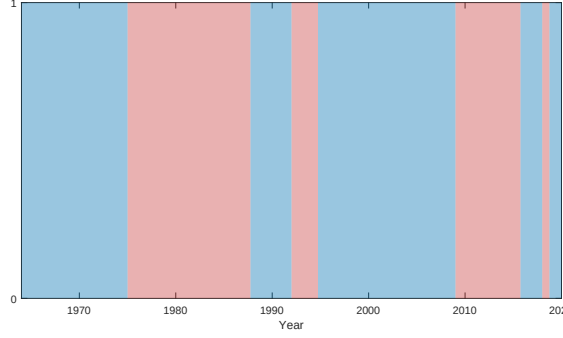


Figure 7: **Fiscal capacity for Social Security.**

Periods of constrained fiscal capacity to finance SS spending, based on the status of the Federal Old-Age and Survivors Insurance trust fund and the Federal Disability Insurance trust fund. Shaded areas reflect the joint status of the trust funds: period of fiscal strain when both funds are in deficit (red areas), versus periods without fiscal strain when at least one fund is in surplus (blue areas).

SS programs at the time of the announcement.

Identifying fiscal capacity for SS spending. The Social Security Act of 1935 established Social Security. This program is administered by the Social Security Administration (SSA) and is primarily funded through payroll taxes. These taxes are allocated to two trust funds: the Federal Old-Age and Survivors Insurance (OASI) and the Federal Disability Insurance (DI). Although these are two legally separate funds, assessments of the system’s financing often consider the combined programs (OASDI). To assess the availability of funds for Social Security, we examine the surplus or deficit in the OASDI trust fund. We assign a value of one if both (OASI and DI) funds are in deficit (red areas in Figure 7), zero otherwise.¹⁵

Results. Figure 8 presents impulse responses to an anticipated shock to Social Security spending for the period 1968q1-2019q4. Similar to Medicare and Medicaid, Social Security spending primarily consists of transfers to individuals; hence, each local projection includes four lags of transfers, as well as four lags each of the shock, output, and the outcome variable. The red (blue) lines indicate periods in which the OASDI trust funds are in deficit (surplus). The transmission of news shocks to SS spending differs from those of healthcare

¹⁵The index takes a value of one to identify the following periods as times of strain for SS spending: 1975-87, 1992-94, 2009-15, 2018. Results are robust if one aggregates the surplus/deficit in the two funds, if one considers primary surplus/deficit or considering just the OASI trust fund. See Appendix A for further details on these funds.

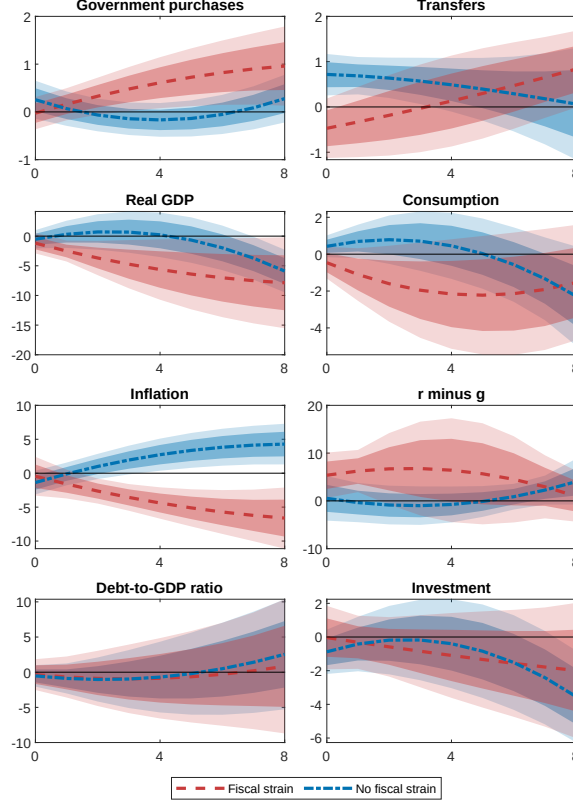


Figure 8: **Impulse responses to anticipated shocks to Social Security, conditional on its fiscal capacity.**

Responses to an anticipated shock to Social Security, conditioning on the periods of fiscal strain (red lines) or absence of fiscal strain (blue lines) in the OASDI funds, as described in Figure 7. Each panel reports impulse responses from state-dependent local projections estimated over the sample 1968q1-2019q4. The set of controls always includes four lags of the shock, output, transfers, and the outcome variable. Shaded areas: 68% (dark) and 90% (light) confidence bands.

programs and National Defense: when Social Security funding is ample, anticipated increases in spending appear to be more inflationary. That said, the responses of real activity, the growth-interest rate differential, and the debt-to-GDP ratio are less definitive, lacking clear evidence to consistently support Ricardian behavior.

How should we interpret these findings? The very nature of SS spending makes it more challenging to assess the extent of fiscal backing. Such programs typically create entitlements that are hard to reverse – benefits that are persistent, politically sensitive, and often seen as permanent commitments. For example, pensions that individuals incorporate into their long-term financial planning cannot easily be reduced or revoked. At the same time, the fiscal capacity to support these entitlements can change over time, influenced by demographic trends, economic growth, and broader macroeconomic conditions.

On the other hand, a more relaxed budget constraint for social spending may lead economic agents to expect more ambitious and sustained expansions in welfare programs, which are more likely to raise concerns about fiscal sustainability in the long run, especially when demographic and economic fundamentals are likely to change. This combination of entitlement rigidity and the inherent difficulty in assessing the long-run fiscal sustainability of such programs heightens concerns about the financial viability of the long-term spending commitments introduced in the 1960s.

During this period, U.S. President Lyndon Johnson launched the Great Society initiatives – an ambitious effort to expand social spending and entitlements over several years. The U.S. economy was experiencing rapid growth, and a favorable demographic structure provided supportive conditions for increased government funding of social programs. Indeed, the OASI and DI trust funds, used here as proxies for financial strain in social spending, were largely in surplus at the time.

To support these interpretations, we present two key findings. First, as Figure 9 shows, the largest SS shocks in our sample occurred during the 1960s and the first half of the 1970s, a period characterized by abundant OASI and DI funds.¹⁶ Second, as Figure 10 shows, if we start the sample from 1975q1, thereby excluding the shocks from 1960s and early 1970s, the effects of SS spending are Ricardian, regardless of the availability of OASI and DI funds. This finding echoes the results for the other major components of mandatory spending, Medicare and Medicaid. Moreover, it suggests that the inflationary effects of anticipated shocks to SS spending are primarily driven by the sizable shocks of the 1960s and early 1970s, a period when the U.S. government launched an ambitious expansion of entitlements.

3.3 Taking Stock

We find substantial heterogeneity in how anticipated changes to spending categories affect inflation and economic activity. National defense – discretionary spending – can have sizable macroeconomic effects, especially during fiscal stress. When perceived as unfunded, that is, not backed by immediate tax increases, it can generate positive wealth effects, raising output

¹⁶See Appendix, Part D, for analogous figures displaying anticipated shocks to Medicare & MM and ND.

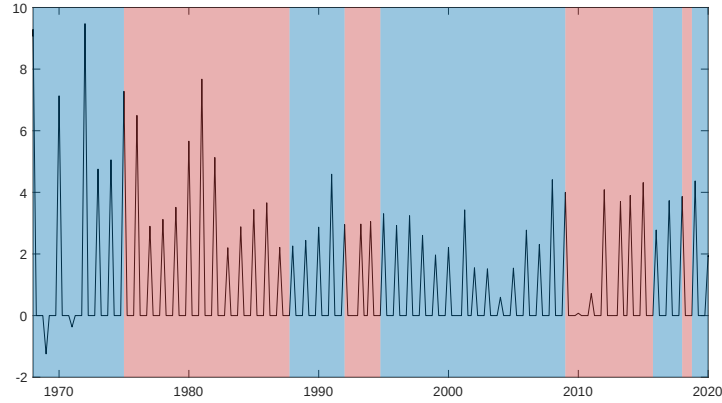


Figure 9: **Anticipated shocks to Social Security and fiscal capacity based on the OASDI funds.**

Black line: time series of anticipated shocks to Social Security, in percentage points. Shaded areas: periods of fiscal strain (red) or absence of fiscal strain (blue) in the OASDI funds, as described in Figure 7.

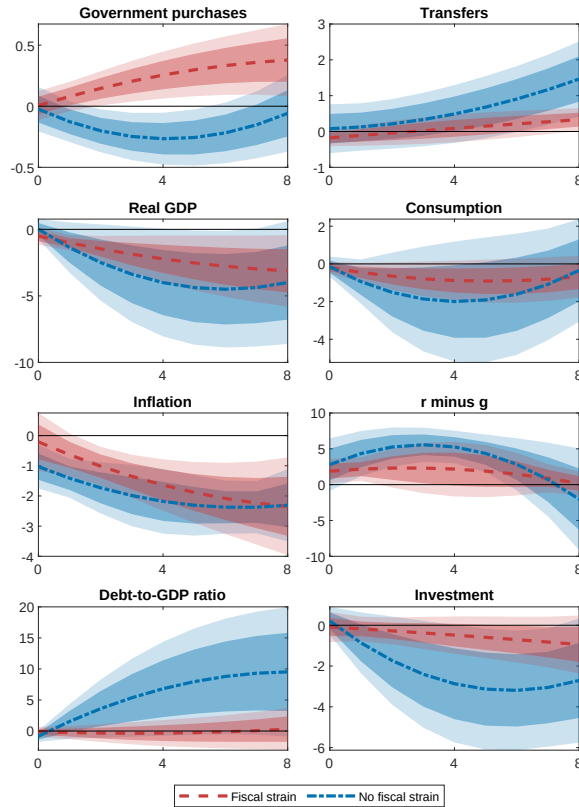


Figure 10: **Impulse responses to anticipated shocks to Social Security (SS), conditional on its fiscal capacity, over the subperiod 1975q1-2019q4.**

Responses to an anticipated shock to Social Security, conditioning on the periods of fiscal strain (red lines) or absence of fiscal strain (blue lines) in the OASDI funds, as described in Figure 7. Each panel reports impulse responses from state-dependent local projections estimated over the sample 1975q1-2019q4. The set of controls always includes four lags of the shock, output, transfers, and the outcome variable. Shaded areas: 68% (dark) and 90% (light) confidence bands.

and inflation. These effects are accommodated by the central bank, which allows real interest rates to fall to help stabilize debt. By contrast, when financed through higher current taxes the effects align with Ricardian predictions, reducing consumption.

For defense spending, the immediate availability of funds can serve as a useful proxy for fiscal backing. Since National defense requires annual congressional approval, when such approval occurs during a period of fiscal strain the credibility of that backing is weakened.

Nevertheless, the near-term availability of funds is not always a reliable predictor of the effects of anticipated changes in spending across all different categories. For instance, spending shocks to Medicare & Medicaid, as components of mandatory spending, consistently exhibit muted macroeconomic effects regardless of the government's near-term fiscal capacity. Possibly because these programs primarily fund healthcare services, rather than direct transfers that raise disposable income, they have limited impact on consumption or inflation expectations. This suggests that, due to their mandatory status, such programs are effectively treated as being always backed fiscally, even when resources are scarce in the short run.

The evidence for Social Security is more nuanced. Our results show that the immediate availability of funds is not, by itself, a reliable proxy for fiscal backing. Unlike Medicare and Medicaid, however, Social Security generates expectations of permanent transfers, which raise consumption and put upward pressure on prices, only in the wake of structural spending shifts. Because such shifts are persistent, their financing coverage extends over time and thus entails greater uncertainty. A prominent example is the substantial expansion of Social Security during the mid-1960s, associated with President Lyndon Johnson's Great Society initiatives. In the case of Social Security, therefore, long-run program characteristics, rather than short-term fund availability, seem to be a critical factor for determining the presence or absence of fiscal backing.

To complement this analysis, Appendix B extends the classification to a broader distinction between mandatory and discretionary spending. We confirm that mandatory spending shocks generally elicit Ricardian responses, irrespective of short-run fiscal strain, while discretionary shocks are more likely to generate non-Ricardian effects, particularly during

periods of limited fiscal space.

A closer look at Defense Spending: ‘Warflation’. In this section, we are going to dig deeper into the inflationary effect of ND spending by focusing on specific historical episodes. This exercise should be read as a validation of the main mechanism we have discussed throughout the paper, namely, the distinction between Ricardian and non-Ricardian responses of ND spending, depending on the fiscal context. Rather than relying on regime indicators or fiscal indices, we now leverage the variation in defense outlays associated with historical U.S. war efforts.

To this end, we adopt an alternative classification of states based on war periods.¹⁷ War-related buildups in defense spending provide a natural experiment to test Ricardian versus non-Ricardian effects, as such spending is typically large and unexpected – driven by external geopolitical needs.

We examine average annual ND spending as a percentage of GDP during each U.S. war period in our sample considering, when defense outlays remained elevated, even mobilization and demobilization quarters. With this definition, average ND spending amounted to 13.4% of GDP during the Korean War (1950Q2–1954Q3), 7.5% during the Vietnam War (1964Q1–1975Q4), and 5.15% during the Gulf War (1990Q1–1991Q4). From 2001 to 2011, during the Iraq and Afghanistan wars, defense spending averaged 4.1%, and in the subsequent period 2011–2021, corresponding to the later years of the Afghanistan war, it averaged 3.4%. Based on the relative size of ND spending during different war periods, we consider yet another binary indicator variable, which takes the value of one during the periods corresponding to the three more expensive wars (i.e., Korean war, Vietnam war and Gulf war) and zero otherwise. This indicator is then used within the local projection framework of equation (2) to estimate state-dependent responses of inflation and other macroeconomic variables.

We find (see Figure 11) that outside these dates (highlighted in blue), impulse responses

¹⁷While the previous analysis of defense spending was restricted to the 1968q1–2019q4 period to ensure comparability with MM spending, in what follows we extend the sample back to 1950q1. This allows us to capture the onset of the Korean War and more accurately study the inflationary dynamics associated with major U.S. military conflicts.

behave as Ricardian, whereas during high-cost war periods (in red), they reflect a non-Ricardian pattern. Anticipated defense shocks lead to a sustained increase in government purchases in both states, with a much stronger response in periods of high war expenditure. In such episodes, defense spending significantly boosts GDP: output rises markedly and remains elevated, whereas it contracts during more ordinary times. Consumption does not increase in either case, and investment—although not statistically significant—performs better in war episodes, while it falls sharply otherwise.

Inflation also shows a clear divergence: it declines in ordinary circumstances but rises significantly when wars are costly, thereby reducing the interest–growth differential ($r - g$). This leads to very different debt dynamics: the debt-to-GDP ratio increases steadily in peacetime conditions, while it remains broadly stable, or even slightly declines, when fiscal pressures are driven by major wars.

These findings suggest that military spending shocks generate large multipliers and strong inflationary pressures, particularly during wartime. This historical episode analysis thus reinforces the main mechanism proposed in the paper: the fiscal context crucially conditions the macroeconomic effects of ND spending, with high-cost wartime periods displaying markedly non-Ricardian dynamics.

4 Conclusions

Fiscal policy is inherently complex, raising questions about whether different categories of spending in the U.S. Federal Budget have a larger impact on output and prices, and whether they have historically received varying levels of fiscal backing. Does the choice of fiscal tools matter to stimulate the economy during a recession? And, importantly, which spending categories pose a greater risk to price stability? This paper shows that not all spending is alike in its macroeconomic impact.

Shifts in discretionary spending, particularly in National Defense, display pronounced non-Ricardian effects, especially during times of limited or sharply reduced funding. In such contexts, insufficient fiscal resources can undermine the credibility of support for military

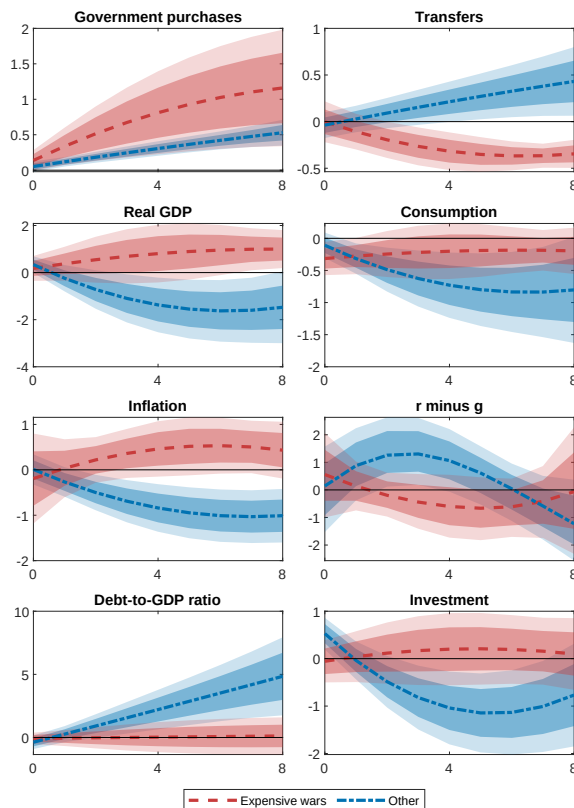


Figure 11: **Impulse responses to anticipated shocks to National Defense, distinguishing between high-cost war periods versus other periods.**

Responses to an anticipated shock to National Defense, distinguishing between relatively more expensive war periods (red lines) and other times (blue lines). We classify as high-cost war periods the Korean war (1950q2-1954q3) with average ND spending at 13.4% of GDP, the Vietnam War (1964q1-1975q4) at 7.5%, and the Gulf War (1990–1991), at 5.15%. For reference, defense spending averaged 4.1% of GDP during the Iraq War and the first years of the Afghanistan War (2001-2011), and 3.4% of GDP in the subsequent years (2011-2021). Each panel reports impulse responses from state-dependent local projections estimated over the sample 1950q1-2019q4. The set of controls always includes six lags of the shock, output, government purchases, and the outcome variable. Shaded areas: 68% (dark) and 90% (light) confidence bands.

expenditures, generating wealth effects that translate into higher output and inflation.

By contrast, changes in mandatory spending, especially in Medicare and Medicaid, generally produce Ricardian effects, irrespective of funding pressures for these programs. Social Security spending presents a more nuanced case: large and sustained expansions of entitlements, such as those enacted during the Great Society era, have historically created expectations of permanent transfers, with meaningful upward pressure on prices.

Overall, these findings suggest that the macroeconomic consequences of fiscal policy depend not only on the size and category of spending, but also on the credibility and persistence of its fiscal backing.

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Appendix

A Data

The outcomes of our local projections are quarterly data from NIPA tables of aggregate federal purchases & investment, aggregate federal transfers, real GDP, real consumption, real investment, inflation, the interest rate-growth differential ($r - g$), and debt over GDP. All variables are expressed in levels. The interest rate-growth differential ($r - g$) is calculated as the difference between the nominal interest rate and the nominal GDP growth rate, both measured at quarterly frequency.

Trust funds for MM Medicare is funded through two main trust funds: The Hospital Insurance (HI) Trust Fund and Supplementary Medical Insurance (SMI) Trust Fund. The Hospital Insurance (HI) Trust Fund is primarily financed through payroll taxes collected under the Federal Insurance Contributions Act (FICA) for employees and employers, and under the Self-Employment Contributions Act (SECA) for self-employed individuals. Employees and employers each contribute a fraction of their wages, while high-income earners pay an additional Medicare surtax. The HI Trust Fund faces solvency challenges, and its balance depends on the balance of incoming payroll taxes and outgoing Medicare Part A expenses. The SMI Fund is primarily funded by general tax revenues and monthly premiums paid by Medicare enrollees and relies heavily on annual appropriations.

Trust funds for SS The Social Security Act mandates that Social Security spending be primarily financed through payroll taxes collected under the Federal Insurance Contributions Act (FICA). The OASI Trust Fund finances benefits for retirees and their survivors. It is primarily supported by payroll taxes levied on employees and employers, with a portion of earnings subject to these taxes capped at a specific threshold. The funds collected are used to pay monthly benefits to eligible retirees and to the survivors of deceased workers. The Disability Insurance (DI) Trust Fund provides benefits to individuals who become disabled and can no longer work. Like the OASI Trust Fund, the DI Trust Fund is also funded through payroll taxes. However, the DI Trust Fund has faced challenges in recent years due to increasing claims and demographic shifts, raising concerns about the sustainability of its funding.

Periods of fiscal strain To assess the availability of funds for Medicare & Medicaid and Social Security, we examine the surplus or deficit in their trust funds. The data for both SS and MM trust funds are drawn from Table 13.1, Cash Income, Outgo, and Balances of the Social Security and Medicare Trust Funds: 1936–2028 in the U.S. Budget.

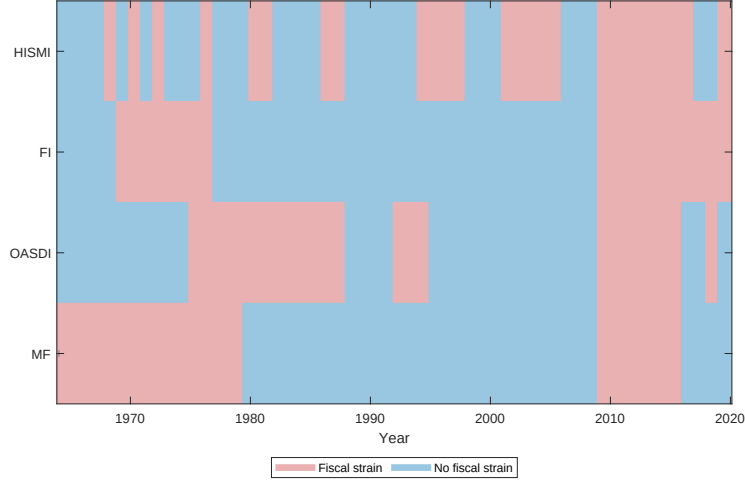


Figure 12: **Comparison of periods implied by alternative fiscal capacity indicators.** The figure contrasts the periods of high and low fiscal capacity identified by alternative indicators, serving as a basis for state-dependent analysis. Red areas: periods of fiscal strain. Blue areas: periods of absence of fiscal strain.

B Mandatory and Discretionary spending

Does the variation in behavior across states appear only when examining specific subcategories of expenditure, or is it also evident in broader, aggregated categories like mandatory or discretionary spending? We divide government expenditure into its two main components – mandatory and discretionary – and examine the impulse responses following an anticipated shock to each, using the longest time span covered by the data (1964q1-2019q4). The specification includes four lags of the control variables, namely the outcome variable, output, and both transfers and public purchases for mandatory expenditures and only public purchases for discretionary spending. We begin by estimating a linear model, as in equation (1). As Figure 13 shows, there are no marked differences in the responses of the variables between the two types of spending.

We employ the FI to distinguish a state with high fiscal capacity (where federal tax revenues exceed the amount required to cover mandatory expenses, with the surplus available to fund discretionary spending) (in red) from a state with a low capacity (in blue), using (2). Unlike in the linear case, mandatory and discretionary spending now exhibit different behaviors. As shown in Figure 14, the impulse response functions for mandatory spending

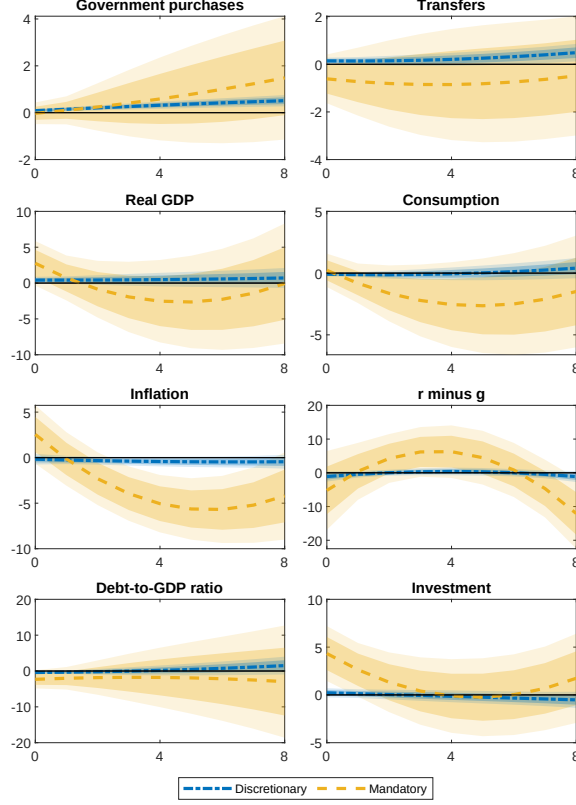


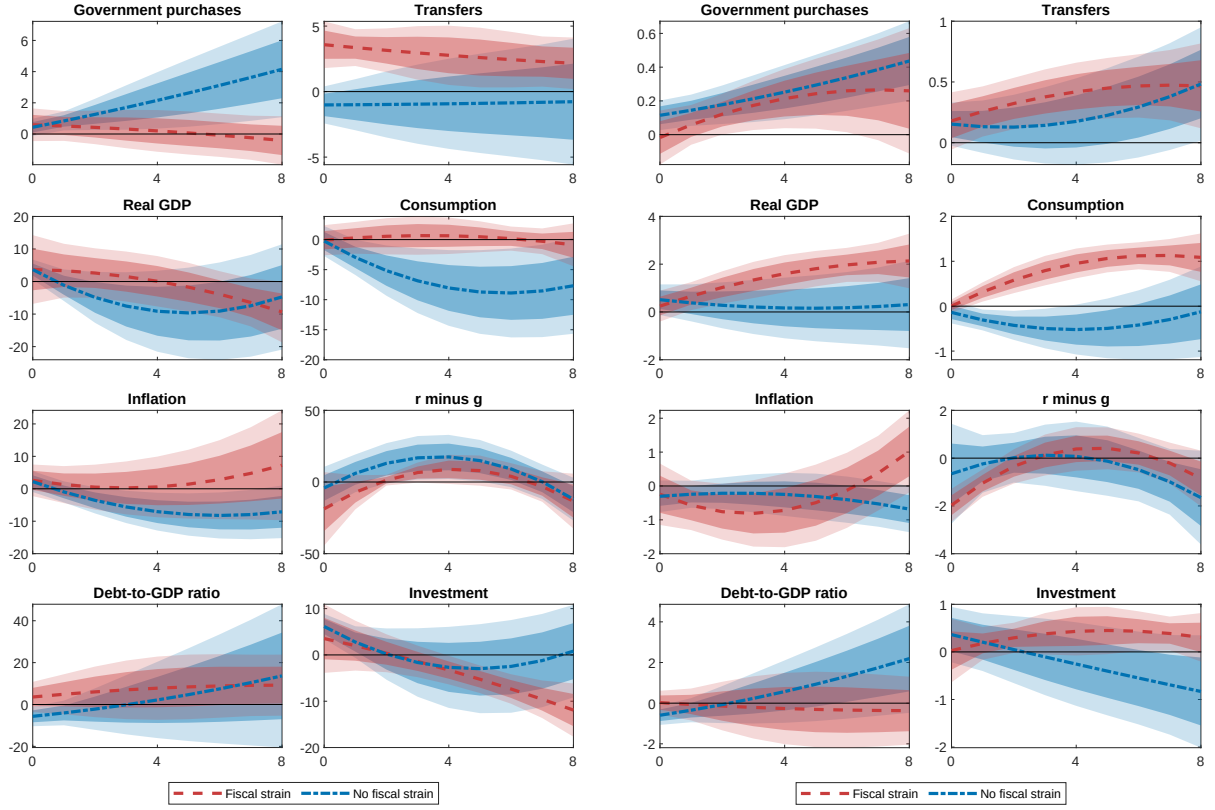
Figure 13: **Effects of anticipated spending shocks to mandatory and discretionary spending.**

Blue lines: responses to an anticipated shock to mandatory spending. Yellow lines: responses to an anticipated shock to discretionary spending. Each panel reports impulse responses from linear local projections estimated over the sample 1964q1-2019q4. The set of controls always includes four lags of the shock, output, and the outcome variable, plus four lags of both transfers and government purchases for mandatory spending and only government purchases for discretionary spending. Shaded areas: 68% (dark) and 90% (light) confidence bands.

consistently follow the pattern of a *funded* spending shock. In contrast, impulse responses for discretionary spending behave like those following *funded* shocks when the FI is high but resemble *unfunded* shocks when the FI is low, most evidently with respect to the response of output and consumption.

The result for mandatory spending is not surprising. PAYGO, applied to mandatory spending to control deficit increases, requires that any rise in direct spending or revenue decrease must be offset by other spending cuts or revenue increases.¹⁸ These mechanisms ensure that such expenditures are consistently funded. Notably, PAYGO does not apply to

¹⁸PAYGO, introduced in 1990 to curb deficits, was waived in the late 1990s due to surpluses and abandoned in 2002 after deficit-financed tax cuts. It was reinstated in 2007 and codified in the 2010 Statutory PAYGO Act, which mandates that new deficit-increasing legislation be offset or trigger automatic spending cuts (sequestration). Social Security and Medicaid are exempt, while Medicare can be reduced by up to 4%.



(a) Anticip. shock to mandatory, FI.

(b) Anticip. shock to discretionary, FI

Figure 14: Impulse responses to anticipated shocks to mandatory and discretionary spending, conditional on the Fiscal index.

Panel (a): responses to anticipated shocks to mandatory spending. Panel (b): Responses to anticipated shocks to discretionary spending. Each panel reports impulse responses from state-dependent local projections estimated over the sample 1964q1-2019q4. Responses are conditioned on the periods of fiscal strain (red lines) or absence of fiscal strain (blue lines) in the Fiscal index, as described in Figure 5b. The set of controls always includes four lags of the shock, output, and the outcome variable plus four lags of both transfers and government purchases for panel (a) and only government purchases for panel (b). Shaded areas: 68% (dark) and 90% (light) confidence bands.

discretionary spending, which is instead subject to annual budget caps and the appropriations process. Overall, the behavior of mandatory and discretionary spending mirrors the results we obtained for their respective sub-components, respectively, MM that is always funded, and ND whose financing depends on fiscal capacity.¹⁹

¹⁹Note that while the distinction in behavior between high and low FI states is neat for ND, it becomes less so when looking at discretionary spending. This is likely because discretionary spending is an aggregate that encompasses various subcategories, whose behavior may not be as clear-cut as that of ND.

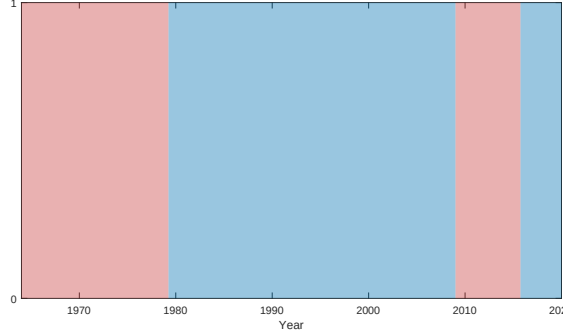


Figure 15: **Monetary and fiscal regimes.**

Periods of unfunded fiscal regime (in red) correspond to the Great Inflation years (1964q1-1979q2) and the years when monetary policy operated at the zero lower bound (2009q1-2015q4). Periods of funded monetary regime (in blue) correspond to the Great Moderation years (1984q1-2008q4).

C Evidence with alternative states

C.1 The Monetary-Fiscal regime (MF)

Instead of relying on the previously fiscal Index, we now turn to a different approach to capture episodes of fiscal strain. Specifically, we now exploit the well-known distinction between a monetary-led regime (regime M) and a fiscally-led regime (regime F) found in the literature on monetary-fiscal interactions (see [Bianchi, 2012](#); [Bianchi and Ilut, 2017](#); [Bianchi and Melosi, 2014, 2017](#); [Chung et al., 2007](#)).²⁰

We assign a value of one to the unfunded fiscal sub-periods: the Great Inflation years (1964q1-1979q2) and the periods when monetary policy operated at the zero lower bound (2009q1-2015q4), while we assign a value of zero to the funded monetary regime: the Great Moderation years (1984q1-2008q4).²¹ Blue (red) areas in Figure 15 identify a fiscal unfunded (monetary funded) regime.

The decision to extend the fiscal regime beyond the Great Inflation years to the post-

²⁰According to [Leeper \(1991\)](#), the monetary regime assumes an active monetary policy, adhering to the Taylor principle, and a passive fiscal policy. In this regime, fiscal expansions are always backed by future fiscal surpluses. Conversely, the fiscal regime assumes a passive monetary policy and an active fiscal policy: here, fiscal expansions are unbacked and generate wealth effects.

²¹While there is a broad consensus on considering the Great Moderation as a monetary regime, the Great Inflation regime is more debated. Papers assuming a constantly passive fiscal policy detect, in the Great Inflation period, a double passive (hence, indeterminate) regime (see [Lubik and Schorfheide, 2004](#)). However, when allowing for a possible switch in fiscal policy, the pre-Volcker era is found to be consistent with a fiscally-led regime (see [Davig and Leeper, 2007, 2011](#); [Sims, 2011](#)), in which the increasing inflation is attributed to a lack of fiscal discipline.

2009 period is motivated by several factors. Firstly, the post-2009 period is characterized by the zero lower bound (ZLB), indicating a passive monetary policy typical of a fiscal regime. Secondly, while pre-2009 changes in debt ceilings were relatively limited, in the post-2009 period, changes in debt ceilings were much more frequent, with different suspensions occurring. This implies that the ability to finance discretionary spending in general, and ND spending in particular, was subject to greater uncertainty and variability. Third, as [Bianchi et al. \(2023\)](#) stress, the amount of unfunded spending accelerates in the aftermath of the Great Recession.

Given that this regime-based indicator is now used in place of the fiscal index, we reassess the baseline results presented in the paper. In particular, we evaluate how the responses to government spending shocks vary under this new specification – both for defense spending and across the discretionary and mandatory components of government expenditures. Figure 16 shows, for the sample 1964q1-2019q4, the responses of the variables to an anticipated shock to Defense spending under the fiscal regime (in blue) and under the monetary regime (in red). Under regime M (F) output, consumption and investment decrease (increase), while the real rate and $(r - g)$ increase (decrease) or are not significant. Inflation decreases (increases) under M (F), increasing (reducing) debt. Therefore, in regime M, the impulses behave as those following an anticipated funded spending shock; in regime F, the impulses behave as those following an anticipated unfunded spending shock. The results we obtain employing the FI are therefore confirmed.

Looking at mandatory and discretionary spending across the MF states, we find that results under FI states are confirmed. As Figure 17 shows, discretionary spending behaves as Ricardian under the Monetary regime and as non-Ricardian under the Fiscal regime. On the other hand, mandatory spending has always a Ricardian behavior both under the Monetary regime and under the Fiscal regime.

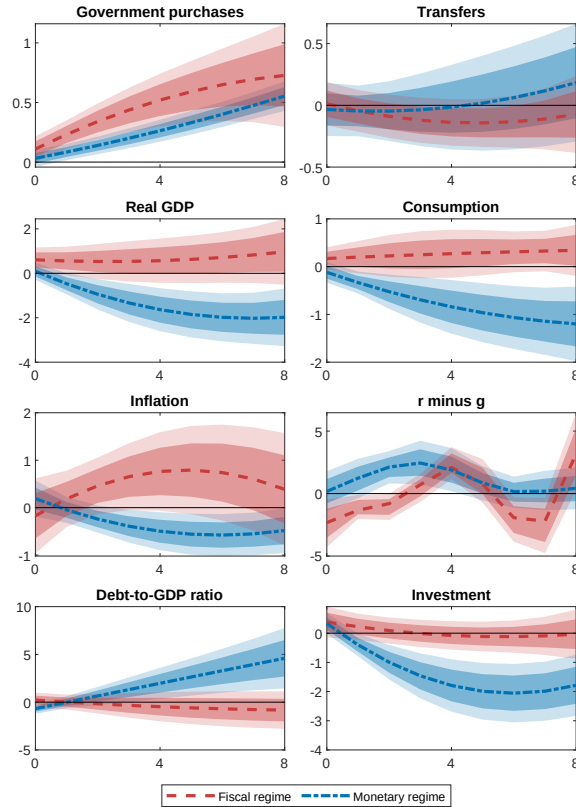
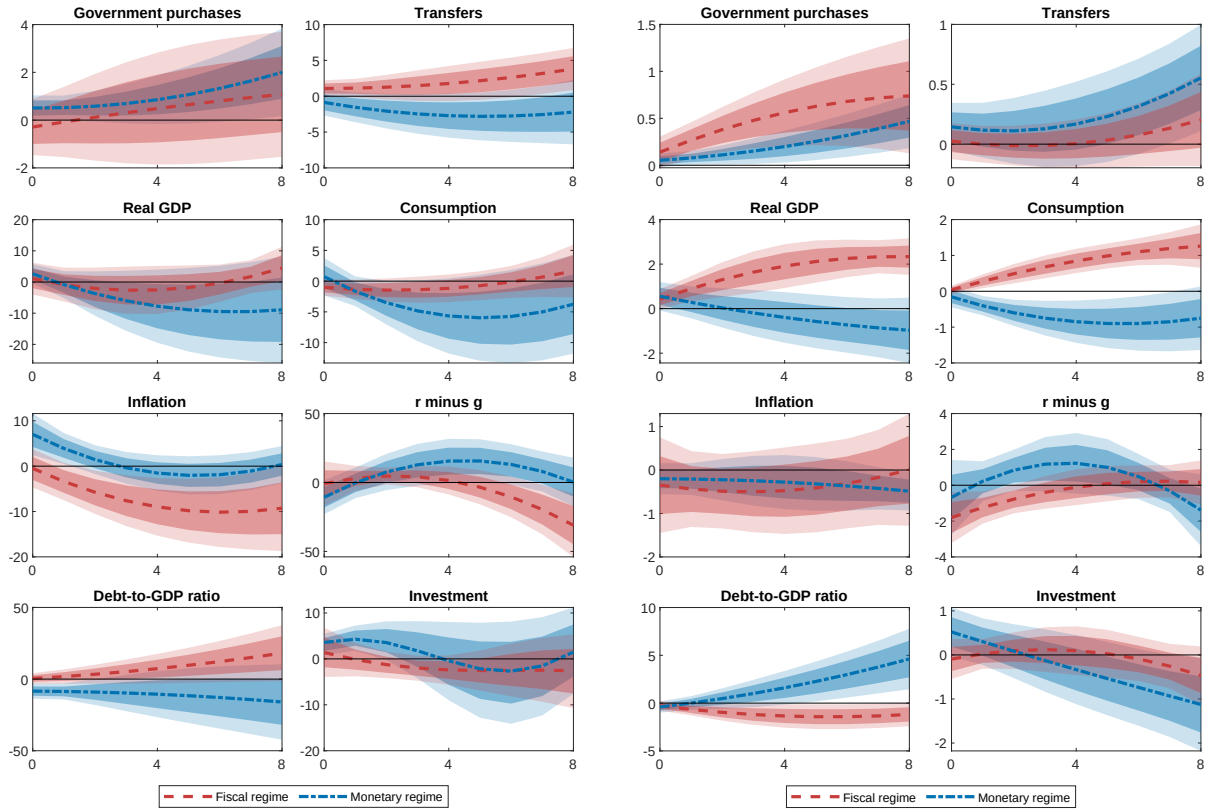


Figure 16: **Impulse responses to anticipated shocks to National Defense, conditional on monetary/fiscal regimes.**

Responses to an anticipated shock to National Defense, conditioning on unfunded fiscal regimes (red lines) or funded monetary regimes (blue lines), as described in Figure 15. Each panel reports impulse responses from state-dependent local projections estimated over the sample 1964q1-2019q4. The set of controls always includes four lags of the shock, output, government purchases, and the outcome variable. Shaded areas: 68% (dark) and 90% (light) confidence bands.



(a) Anticip. shock to mandatory, M/F regimes (b) Anticip. shock to discretionary, M/F regimes

Figure 17: Impulse responses to anticipated shocks to mandatory and discretionary spending, conditional on the monetary/fiscal regimes.

Panel (a): responses to anticipated shocks to mandatory spending. Panel (b): Responses to anticipated shocks to discretionary spending. Each panel reports impulse responses from state-dependent local projections estimated over the sample 1964q1-2019q4. Responses are conditioned on the unfunded fiscal regimes (red lines) or funded monetary regimes (blue lines), as described in Figure 14. The set of controls always includes four lags of the shock, output, and the outcome variable plus four lags of both transfers and government purchases for panel (a) and only government purchases for panel (b). Shaded areas: 68% (dark) and 90% (light) confidence bands.

D Purged shock measures

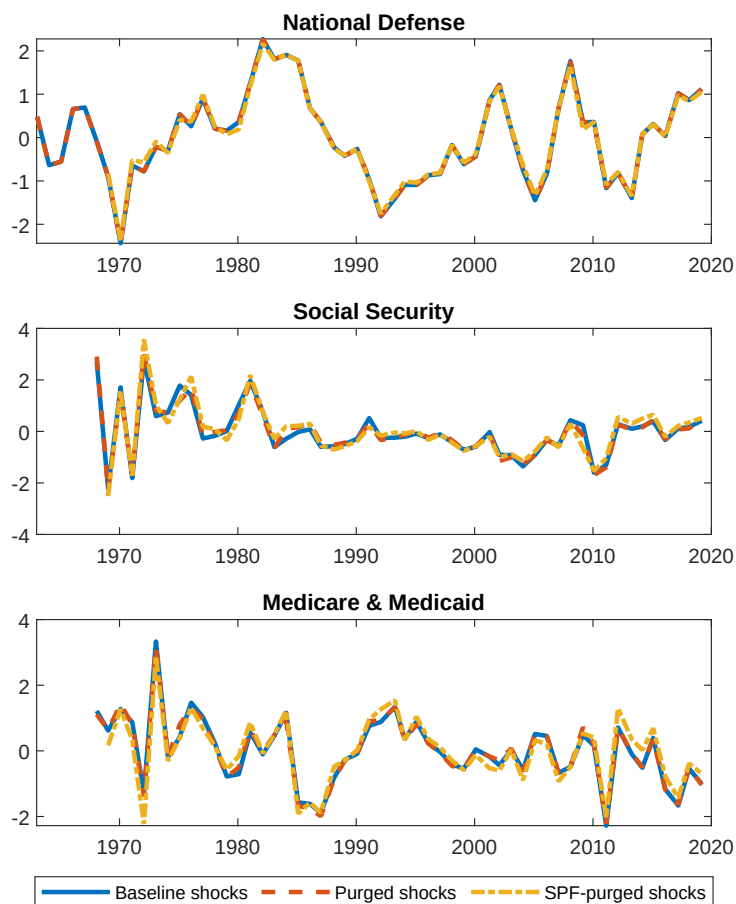
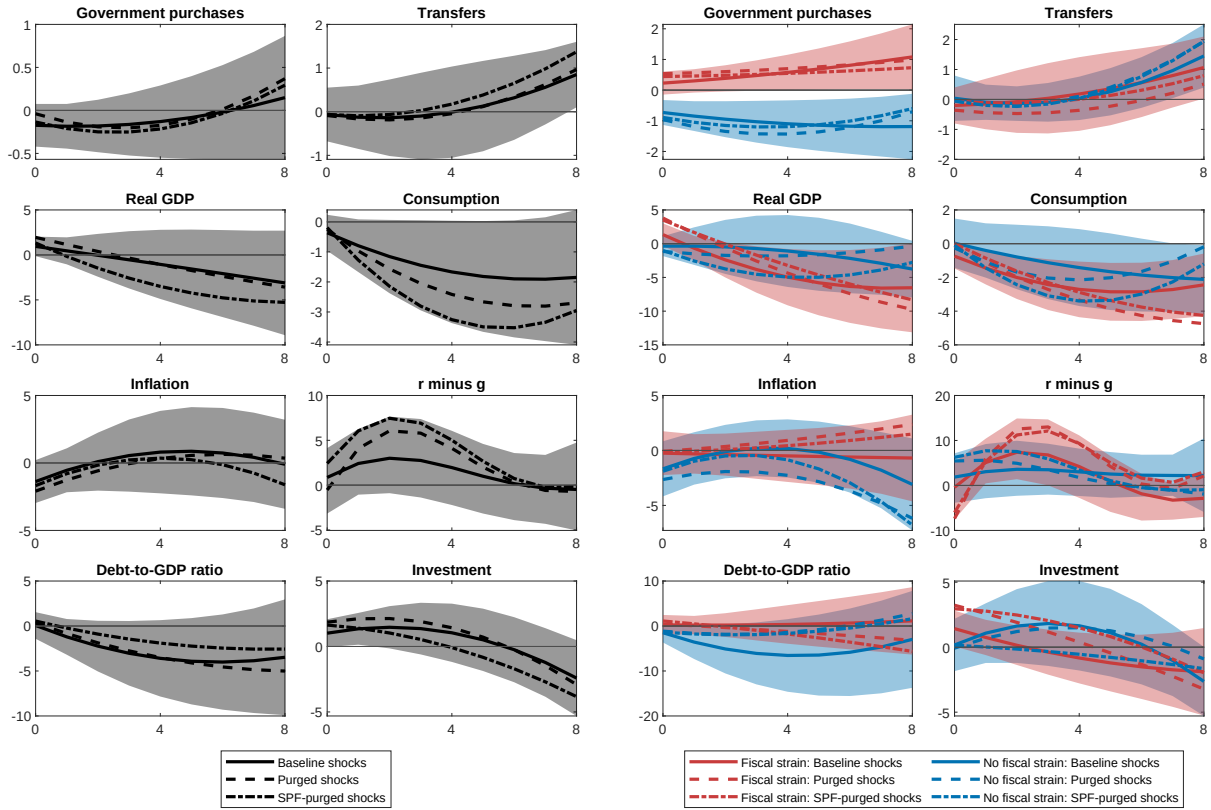


Figure 18: **Baseline vs. purged measures of shocks.**

The blue solid lines show the baseline measure of anticipated government spending shocks, constructed from the spending projections in U.S. budgets for each spending subcategory. The red dashed lines represent purged shocks obtained by regressing the baseline shocks on a constant and the expected growth rate of real GDP reported in the budgets. The yellow dash-dotted lines represent purged shocks obtained by regressing the baseline shocks on a constant, the expected growth rate of GDP, and the expected change in unemployment reported in the Survey of Professional Forecasters.

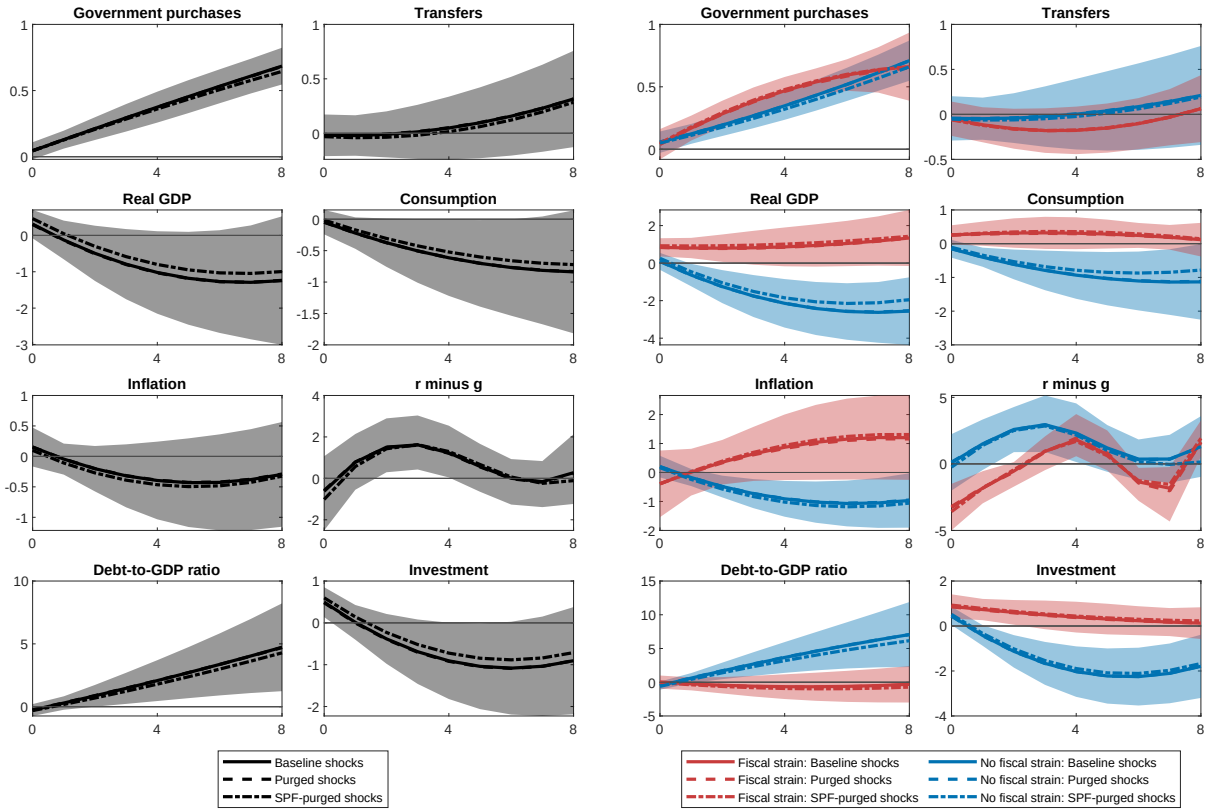


(a) Anticip. shock to MM, linear

(b) Anticip. shock to MM, HISMI

Figure 19: Impulse responses to anticipated shocks to Medicare and Medicaid: Baseline vs. purged shocks.

Panel (a): linear responses to an anticipated shock to Medicare & Medicaid. Panel (b): state-dependent responses to an anticipated shock to Medicare & Medicaid, conditioning on the periods of fiscal strain (red lines) or absence of fiscal strain (blue lines) in the HISMI funds. In all panels, the solid lines show the responses to the baseline shocks, the dashed lines to the purged shocks, and the dash-dotted lines to the SPF-purged shocks. All responses are estimated over the sample 1968q1-2019q4. The set of controls always includes four lags of the shock, output, and transfers. Shaded areas: 90% confidence bands for the baseline shocks.

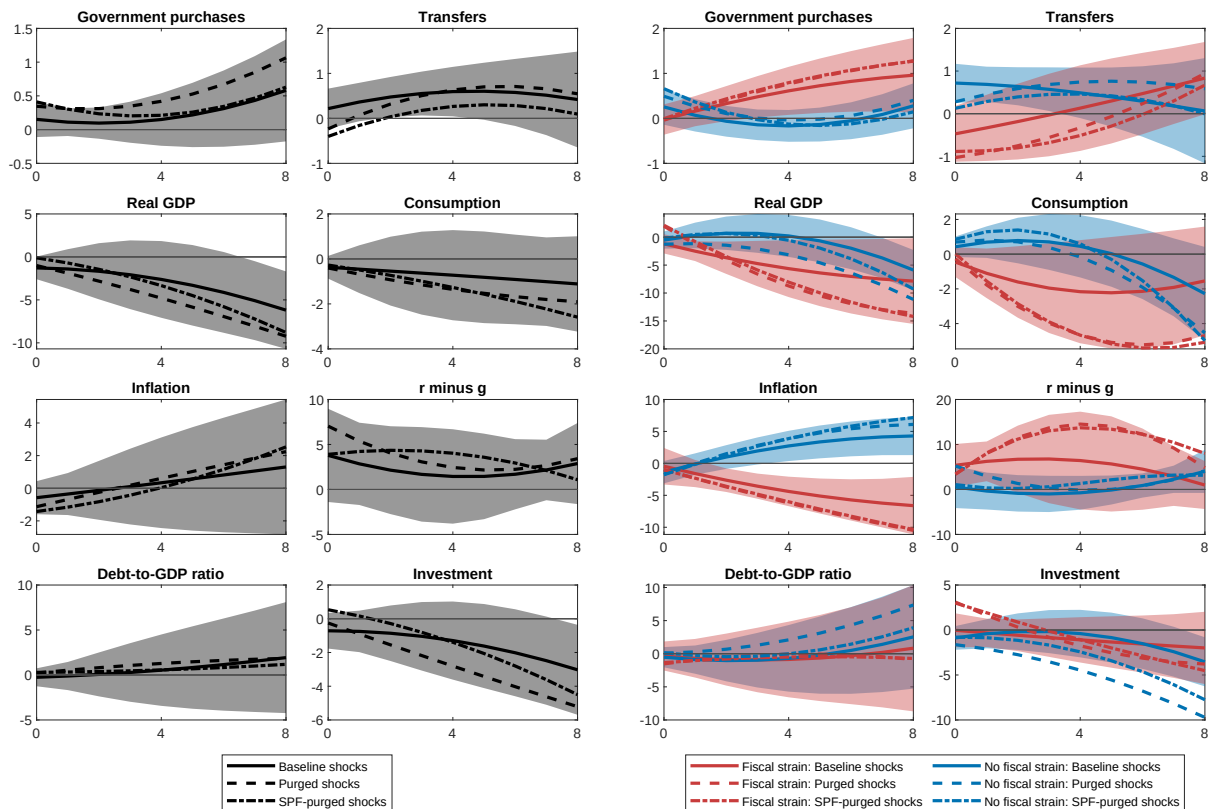


(a) Anticip. shock to ND, linear

(b) Anticip. shock to ND, FI

Figure 20: Impulse responses to anticipated shocks to National Defense: Baseline vs. purged shocks.

Panel (a): linear responses to an anticipated shock to National Defense. Panel (b): state-dependent responses to an anticipated shock to National Defense, conditioning on the periods of fiscal strain (red lines) or absence of fiscal strain (blue lines) in the Fiscal index. In all panels, the solid lines show the responses to the baseline shocks, the dashed lines to the purged shocks, and the dash-dotted lines to the SPF-purged shocks. All responses are estimated over the sample 1968q1-2019q4. The set of controls always includes four lags of the shock, output, and government purchases. Shaded areas: 90% confidence bands for the baseline shocks.



(a) Anticip. shock to SS, linear

(b) Anticip. shock to SS, OASDI

Figure 21: Impulse responses to anticipated shocks to Social Security: Baseline vs. purged shocks.

Panel (a): linear responses to an anticipated shock to Social Security. Panel (b): state-dependent responses to an anticipated shock to Social Security, conditioning on the periods of fiscal strain (red lines) or absence of fiscal strain (blue lines) in the OASDI funds. In all panels, the solid lines show the responses to the baseline shocks, the dashed lines to the purged shocks, and the dash-dotted lines to the SPF-purged shocks. All responses are estimated over the sample 1968q1-2019q4. The set of controls always includes four lags of the shock, output, and transfers. Shaded areas: 90% confidence bands for the baseline shocks.

E Supplementary Figures

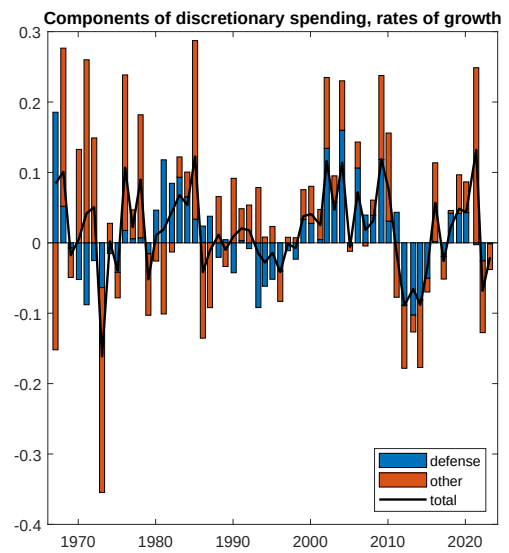
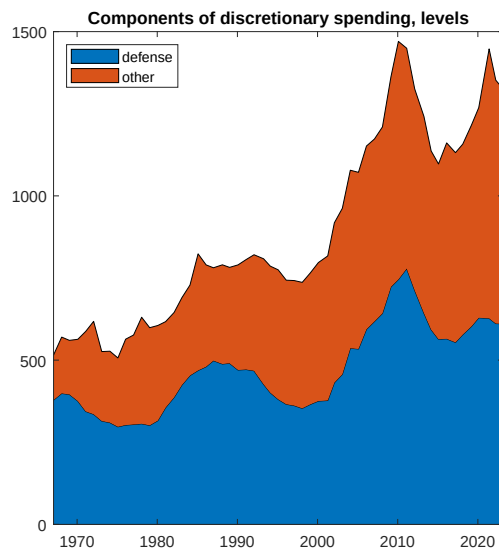
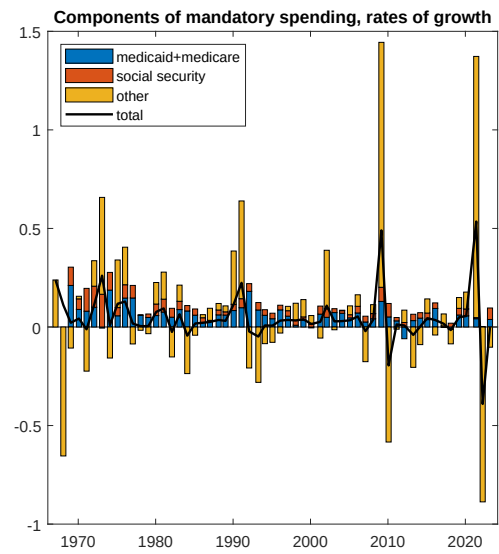
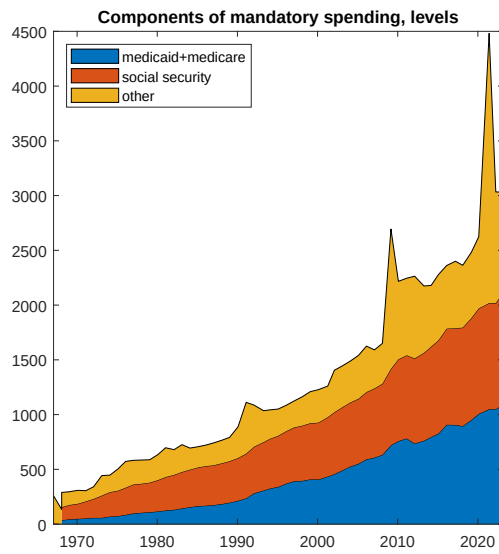


Table 1: Historical debt ceiling levels

Date	Debt Ceiling	Change in Debt Ceiling
August 5, 1997	5,95	450
June 11, 2002	6,4	450
May 27, 2003	7,384	984
November 16, 2004	8,184	800
March 20, 2006	8,965	781
September 29, 2007	9,815	850
June 5, 2008	10,615	800
October 3, 2008	11,315	700
February 17, 2009	12,104	789
December 24, 2009	12,394	290
February 12, 2010	14,294	1,900
January 30, 2012	16,394	2,100
February 4, 2013	Suspended	-
May 19, 2013	16,699	305
October 17, 2013	Suspended	-
February 7, 2014	17,212	213
March 15, 2015	18,113	901
October 30, 2015	Suspended	-
March 15, 2017	19,847 (de facto)	1,734
September 30, 2017	Suspended	-
March 1, 2019	22,030 (de facto)	2,183
August 2, 2019	Suspended	-
July 31, 2021	28,500 (de facto)	6,470
October 14, 2021	28,9	400
December 16, 2021	31,4	2,500

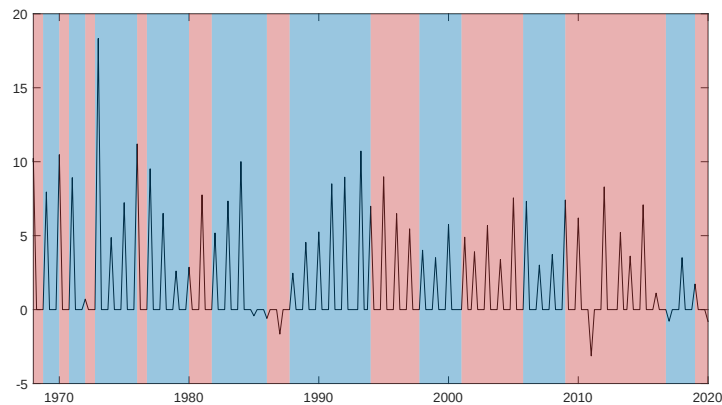


Figure 22: Anticipated shocks to Medicare & Medicaid and fiscal capacity based on the HISMI funds.

Black line: time series of anticipated shocks to Medicare & Medicaid, in percentage points. Shaded areas: periods of fiscal strain (red) or absence of fiscal strain (blue) in the HISMI funds.

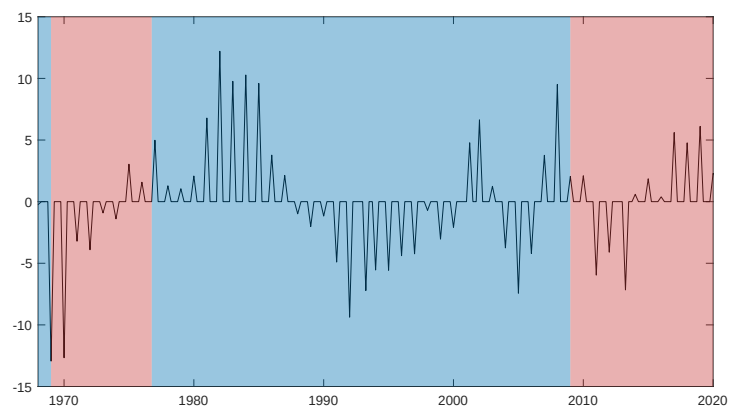


Figure 23: **Anticipated shocks to National Defense and fiscal capacity based on the Fiscal index.**

Black line: time series of anticipated shocks to National Defense, in percentage points. Shaded areas: periods of fiscal strain (red) or absence of fiscal strain (blue) in the Fiscal index.